

Where Shared Explanatory Compression Ceases to Discriminate: A Conditional Phenomenal-Contrast Quotient for AI Consciousness

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Abstract

Assessments of AI consciousness sometimes credit a phenomenal interpretation with all the explanatory compression achieved by a model of architecture, access, self-monitoring, reporting, action control, or hidden predictive structure. This manuscript argues that such credit is warranted only relative to an explicit rival. A whole theory is represented provisionally as $T = (M, B)$, where M contains third-person generative commitments and B maps specified implementations or histories to possibly probabilistic phenomenal assignments. Because this factorization is not unique, the primary object is the whole theory: all observable consequences, bridge-conditioned priors, substrate restrictions, and phenomenal causal effects are included in its induced data measures. Refactorizations that preserve those measures and the phenomenal assignment preserve the audit verdict.

A Phenomenal-Contrast Quotient groups independently admissible theories by equality of their induced probability measures in a preregistered evidence language. The construction is generic empirical-equivalence bookkeeping, not by itself a localization of the philosophical hard problem. It becomes relevant to consciousness only when an empirical class contains serious, target-preserving rivals with different phenomenal laws. Admissibility therefore requires typed consistency, fixed scope, non-post-hoc bridge principles, complete disclosure of observable implications, and genuine phenomenal disagreement rather than a renamed latent label. These conditions constrain but do not settle the disputed coherence of conscious–nonconscious functional duplicates.

The central formal result is correspondingly modest. Empirically equivalent theories have Bayes factor one for every positive-probability event and almost everywhere for likelihood densities relative to a common dominating measure. Their ideal differential data codelength is zero under a fixed coding convention. A shared-mechanism corollary shows that compression achieved equally by two bridge rivals cannot selectively confirm one. A worked comparison between organizational sufficiency and biological-substrate necessity illustrates both the use and the unresolved metaphysical premise of the framework.

Randomized and route-isolated mediated-control ranks and recovery-linked quotient observability are retained as speculative functional diagnostics. They can separate report from same-cut control and behavior-hidden predictive capacity from newly generated capacity, but they are not phenomenal scales. The framework also distinguishes evidence from decision: precaution may lower a welfare-policy threshold without changing the likelihood or posterior probability of consciousness.

Keywords: AI consciousness; phenomenal consciousness; empirical equivalence; explanatory compression; Bayes factor; minimum description length; functionalism; causal intervention; observability; moral uncertainty

Research Question and Hypothesis

Research question

When a model compactly explains an AI system's architecture, access, monitoring, reports, action control, and diachronic hidden capacities, under what conditions does that success discriminate a phenomenal interpretation from a serious nonphenomenal or substrate-restricted rival?

The question is contrastive. It is not answered by showing that a system is recurrent, integrated, globally available, self-representing, highly capable, or difficult to interpret. It requires specification of:

1. the phenomenal proposition;
2. the rival theory;
3. the candidate subject boundary and temporal grain;
4. the admissible observations and interventions;
5. all theory-conditioned parameter priors and substrate restrictions; and
6. the bridge principle that connects third-person structure to phenomenality.

Four claims must remain distinct:

- **Empirical equivalence:** two theories induce the same probability measures in a declared evidence language.
- **External-observer underdetermination:** an assessor using that language cannot update between the theories by likelihood alone.
- **Metaphysical distinctness:** the phenomenal and functional facts could genuinely differ.
- **Explanatory failure:** a functional or physical account fails to explain why experience exists or has its character.

Only the first follows from the formal equivalence definition. The second follows conditionally for an assessor restricted to the registered evidence. The third and fourth require further arguments about phenomenal concepts, identity, realization, explanation, and the admissibility of the rivals.

Hypothesis: conditional bridge non-discrimination

Let T and T' be admissible whole theories. For each experiment e in a fixed evidence language $\mathcal{E}$, let P_T^e and $P_{T'}^e$ be their marginal probability measures after nuisance parameters have been integrated using the priors specified by the respective theories. If

$$P_T^e = P_{T'}^e \quad \text{for every } e \in \mathcal{E},$$

then the data supplied by $\mathcal{E}$ have a unit Bayes factor between them. More precisely:

- for every measurable event A with positive common probability,

$$\frac{P_T^e(A)}{P_{T'}^e(A)} = 1;$$

- if both measures have densities p_T^e and $p_{T'}^e$ relative to a fixed common dominating measure μ_e , then

$$p_T^e(d) = p_{T'}^e(d)$$

for μ_e -almost every d , and the likelihood ratio is one wherever the common density is positive; and

- under a common discrete, quantized, or otherwise fixed coding convention,

$$-\log_2 p_T^e(d) + \log_2 p_{T'}^e(d) = 0$$

almost everywhere.

The result applies to any empirically equivalent theories. The consciousness-specific corollary is that, if

$$T_C = (M, B_C) \quad \text{and} \quad T_0 = (M, B_0)$$

are independently admissible phenomenal rivals with the same induced data measures, then the mechanistic fit shared by M does not differentially confirm B_C over B_0 .

This hypothesis does not establish that the pair exists. Analytic functionalists may deny the coherence of B_0 ; biological theorists may deny the sufficiency asserted by B_C ; illusionists may reject the phenomenal target; and theories of causally efficacious phenomenality may deny empirical equivalence. The framework is intended to identify where those substantive commitments enter, not to settle them by notation.

Evidential standard

The analysis follows five constraints.

1. **Whole-theory comparison.** Bayes factors compare the complete marginal data measures of theories, not verbal labels attached to a shared model.
2. **Preregistration of scope.** The rival, evidence language, boundary, temporal grain, intervention family, and equivalence margin must be fixed before inspecting the target outcomes.
3. **Constitutive restraint.** Evidence for reports, control, prediction, or architecture counts as evidence for phenomenality only through an explicit identity, realization, or psychophysical premise.
4. **No circular validation.** Reports may contribute to consciousness research, but the same report feature cannot define a phenomenal target and then be reused without a measurement model as independent validation of that target.
5. **Source restraint.** The Reyes, Quade, Centered Closure Theory, and Multiscale Reflexive Causal Geometry documents are AI-generated speculative drafts. They are treated as sources of candidate definitions and countermodels, not as authorities or empirical findings ^{[1] [2] [26] [27]}.

Specialist Framework and Contribution

Philosophy of mind fixes the target before measurement

The principal specialist contribution is to preserve distinctions among propositions that can share indicators while having different truth conditions. The access–phenomenal distinction has received explicit empirical and abductive analysis, although that literature does not by itself resolve the contested overflow or sufficiency debates^[30].

Target	Claim	Directly relevant evidence	What the evidence does not establish alone
Phenomenal consciousness	There is something it is like for a candidate subject, or some state has phenomenal character or valence.	For the subject, putative first-person warrant; for an external assessor, public testimony and other data interpreted through a measurement model and bridge principle.	Access, report, causal control, complexity, intelligence, or attribution without that bridge.
Access	Information is available for reasoning, memory, cross-task use, or flexible selection.	Generalization, retrieval, working-memory, routing, and transfer assays.	That the available information is experienced.
Current control	A state affects action at a specified commitment cut, possibly through a designated route.	Registered interventions and causal-channel estimation.	That the causally effective state is experienced or that experience requires motor control.
Self-monitoring	One internal variable tracks another condition such as uncertainty, memory, identity, or authorization.	Calibration and state-to-monitor interventions.	Same-cut use, phenomenality, or truthful introspective testimony.
Self-report	The system emits a sentence or signal about an alleged internal state.	Transcript distributions and operational report accuracy.	That the report is caused by, or testifies to, experience.
Perceived consciousness	Observers attribute consciousness to the system.	Rating experiments and observer-response models.	That the attribution is true.
Agency	The system exhibits temporally extended, boundary-relative, policy-coherent control.	Planning, goal maintenance, correction, tool use, and counterfactual robustness.	Phenomenality or moral patienthood.
Moral status	The system merits protection, rights, or welfare consideration.	Empirical uncertainty plus normative premises about sentence, agency, relations, harm, or precaution.	A descriptive consciousness probability without those normative premises.

These definitions permit conceptual and model-based dissociations. They do not assert that every listed dissociation has been clinically or computationally demonstrated.

First-person evidence must also be disaggregated. A subject's warrant for its own experience is not identical to its public testimony, and public testimony is not identical to an assessor's interpretation of that testimony. For an external AI assessor, only public outputs and their interpretation enter the dataset. Whether they transmit first-person warrant is itself part of the disputed bridge or measurement model.

Registered comparison and admissible theory space

Before target data are examined, an application should register

$$\mathcal{A} = (\mathcal{C}, \mathcal{E}, \mathcal{Y}_\Phi, \mathcal{P}_\Phi, b, \tau, s, \equiv_\Phi),$$

where:

- $\mathcal{C}$ is a family of comparison contexts or candidate histories;
- $\mathcal{E}$ is the family of third-person experiments;
- $\mathcal{Y}_\Phi$ is a phenomenal state space containing a designated null state $\perp_\Phi$;
- $\mathcal{P}_\Phi$ is the set of phenomenal propositions to be compared;
- b is the candidate subject boundary;
- τ is the temporal grain;
- s states the implementation or substrate scope; and
- $\equiv_\Phi$ identifies phenomenal descriptions that differ only by an agreed relabeling.

A whole theory T supplies two typed objects:

1. for every $e \in \mathcal{E}$, a probability measure P_T^e over its third-person outcome space; and
2. a phenomenal kernel

$$\text{Phen}_T(\cdot \mid c) \in \mathcal{P}(\mathcal{Y}_\Phi) \quad (c \in \mathcal{C}),$$

or an equivalent assignment of probabilities to the registered phenomenal propositions.

A deterministic null bridge assigns probability one to $\perp_\Phi$. A deterministic consciousness bridge assigns probability one to a non-null phenomenal state in at least one registered context. Probabilistic bridges are represented directly by their probability laws; no arbitrary probability threshold is needed to define disagreement.

The admissible theory space $\mathcal{T}_\mathcal{A}^{\text{adm}}$ is fixed independently of the realized target dataset. A candidate theory is admitted only if it satisfies the following procedural criteria.

1. **Typed consistency.** Its data measures and phenomenal kernel are mathematically defined on the registered spaces.
2. **Logical coherence relative to its own commitments.** The theory does not both assert and deny the same phenomenal or empirical proposition. This is a minimal condition, not proof of metaphysical possibility.
3. **Scope precision.** Boundary, temporal grain, substrate domain, and transfer class are specified.
4. **Non-post-hoc articulation.** The bridge is stated as a rule over a class of implementations or histories rather than as an opposite label attached only after seeing the target output.
5. **Empirical closure.** All observable restrictions generated by substrate assumptions, phenomenal causal laws, and bridge-conditioned parameter priors are included in P_T^e .
6. **Target preservation.** Rival theories disagree about the registered phenomenal proposition, not merely about an aesthetic, occult, or terminological label.
7. **Nonredundancy.** The phenomenal kernels differ after quotienting by $\equiv_\Phi$.

8. **Calibration disclosure.** Any human, animal, or AI reports used to constrain the bridge are accompanied by a stated measurement model, sampling relation, and transfer assumption.

These rules prevent the most trivial manufacture of a residual fiber. They do not give a neutral criterion of metaphysical seriousness. A functionalist and a biological theorist can still disagree about whether the other's bridge is coherent. That unresolved issue is substantive rather than probabilistic.

Mechanism–bridge factorization and its invariance

For exposition, a theory may be written as

$$T = (M, B).$$

A functional-mechanistic component can be represented as

$$M = (b, \tau, s, \mathcal{X}, \mathcal{U}, \mathcal{O}, K, \Theta),$$

where $\mathcal{X}$ is a state space, $\mathcal{U}$ an intervention family, $\mathcal{O}$ an observation family, K the transition and observation kernels, and Θ the parameter and nuisance structure. A bridge B maps registered implementation histories to phenomenal laws.

This split is not ontologically unique. A substrate condition, prior, psychophysical regularity, or phenomenal causal effect can be written in either column. The audit therefore uses a language-relative normal form:

- $M_{\mathcal{E}}[T]$ contains every commitment of T that affects a third-person probability measure in $\mathcal{E}$, including bridge-induced effects and parameter priors;
- $B_{\mathcal{E}}[T]$ contains the residual phenomenal assignment and its transfer claim after those observable consequences have been included in the empirical core.

This normalization is relative to $\mathcal{E}$. Enlarging the evidence language may move a previously residual commitment into the empirical core.

Two factorizations (M, B) and (M', B') count as permissible refactorizations of the same whole theory when they preserve:

$$P_{(M,B)}^e = P_{(M',B')}^e \quad \text{for every } e \in \mathcal{E},$$

and

$$\text{Phen}_{(M,B)} \equiv_{\Phi} \text{Phen}_{(M',B')} .$$

The principal ledger verdict depends only on whole-theory likelihood ratios and phenomenal disagreement and is therefore invariant under such refactorization. A particular decomposition of model-code length into $\ell(M)$ and $\ell(B \mid M)$ is not invariant; it is meaningful only after a coding convention and normal form have been fixed.

Phenomenal-Contrast Quotient

An experiment $e \in \mathcal{E}$ specifies a regime or adaptive policy, an observation space, and its measurable events. Define

$$T \sim_{\mathcal{E}} T' \iff P_T^e = P_{T'}^e \text{ for every } e \in \mathcal{E}.$$

The empirical quotient is

$$\Omega_{\mathcal{E}} = \mathcal{T}_{\mathcal{A}}^{\text{adm}} / \sim_{\mathcal{E}}.$$

For an empirical class $q \in \Omega_{\mathcal{E}}$, define its phenomenal image by

$$\Phi_{\mathcal{E}}(q) = \{[\text{Phen}_T]_{\equiv_{\Phi}} : T \in q\}.$$

For a fixed proposition $C \in \mathcal{P}_{\Phi}$ and context c , a more directly interpretable image is

$$\Theta_{\mathcal{E}, C, c}(q) = \{P_T(C \mid c) : T \in q\}.$$

The class is **phenomenally heterogeneous** when its phenomenal image contains more than one assignment. A strong conscious–null contrast occurs when, for some C, c ,

$$\{0, 1\} \subseteq \Theta_{\mathcal{E}, C, c}(q).$$

This construction should not be called a proof of the hard problem. It establishes only empirical equivalence plus phenomenal disagreement within the admitted theory space. Its relevance to the philosophical hard problem is conditional: if the rivals are coherent accounts of the same physical or functional totality, then the quotient identifies where the registered third-person likelihood ceases to select between their bridge principles. It does not show that phenomenal facts are metaphysically independent, that reductive explanation is impossible, or that no richer evidence language could matter.

Logical, statistical, and feasible discrimination

The binary horizontal–vertical terminology is insufficient unless three levels are separated.

- **Logical separability:** $P_T^e \neq P_{T'}^e$ for some admissible experiment.
- **Statistical identifiability:** the relevant probability distributions or parameters can be recovered, at least asymptotically, under the experimental design.
- **Feasible discrimination:** a finite registered test reaches specified type-I error, power, robustness, and resource requirements.

Distinct measures may differ only under infeasible interventions or by effects too small to detect with available data. “Different quotient classes” therefore means logical separability, not practical detectability.

Bridge-scope ledger

Every inference to AI phenomenality should disclose:

Entry	Required question
Target	Phenomenality, access, report, control, attribution, agency, or policy?
Rival	Different mechanism, different substrate, different bridge, observer model, or null phenomenal assignment?

Entry	Required question
Admissibility	Why are both theories coherent, non-post-hoc, and target-preserving?
Boundary b	Which model instances, memories, tools, sensors, users, and external stores belong to the candidate subject?
Temporal grain τ	Instantaneous state, commitment cut, temporally thick process, or diachronic system?
Experiments $\mathcal{E}$	Which fixed and adaptive regimes are included?
Observations	Reports only, or also telemetry, weights, memory, hardware, and longitudinal responses?
Substrate scope s	Biological systems, digital systems, or a narrower implementation class?
Nuisance integration	Which theory-conditioned parameter priors generate the marginal likelihood?
Anchor evidence	How are calibration cases measured, and is the same indicator being circularly reused?
Transfer	Why should a bridge calibrated in one substrate or population apply to the target AI?
Coding and priors	Which contributions come from data likelihood, total theory code, or prior commitment?

Two meanings of compression

An AI can internally compress inputs or construct compact predictive representations. Bennett argues that highly compressed representations may become difficult for humans to interpret or communicate with ^[23]. This concerns the system's representation or signal.

A researcher performs a different compression by constructing a probabilistic model that assigns a short code to observed data. The supplied bibliographic record titled *Automated Compression Theory: Explanatory Optimism about the Hard Problem of Consciousness* indicates a direct connection between compression and hard-problem optimism, but the stored record contains no abstract or argument; no more specific interpretation is attributed here ^[28].

Neither form of compression entails phenomenality:

$$C_{\text{internal}} \not\Rightarrow C,$$

and

$$L_D(D \mid M) \ll L_D(D \mid M_{\text{base}}) \not\Rightarrow C,$$

unless a bridge identifies the compressed organization with, or treats it as sufficient for, experience.

Evidence and Literature Synthesis

The standing sufficiency dispute

The supplied bibliography records established discussions under the headings of the hard problem, phenomenal consciousness, functionalism, and the sufficiency claim [29] [31] [34] [35] [36] [37]. Most of these stored records contain titles and bibliographic metadata rather than arguments. They therefore establish the relevance and longevity of the conceptual dispute but do not support a detailed reconstruction or resolution of it.

The present framework does not infer a phenomenal–functional gap from empirical equivalence. It instead asks what follows *if* a sufficiency dispute is represented by independently articulated rival theories. That conditional form leaves open whether analytic functionalism eliminates the nonphenomenal rival, whether biological realization is constitutively necessary, or whether the phenomenal target should be revised.

Theory-derived indicators and architectural success

A prominent AI-consciousness research program derives computational indicators from scientific theories of consciousness. The supplied report applies those indicators to AI systems, gives a negative assessment of the systems considered, and emphasizes that indicator satisfaction would not by itself settle phenomenal consciousness [9]. This evidential restraint is appropriate. An indicator can support a claim that an AI instantiates a theory-specified organization while leaving the phenomenal sufficiency of that organization dependent on a bridge.

Schwitzgebel's survey emphasizes severe theory-relative uncertainty: near-future systems may satisfy some influential theories but not others, weakening simple positive and negative arguments [5]. Multiple indicators should not be counted as independent confirmations merely because they converge verbally. Their evidential contribution depends on a joint measurement model that represents shared causes, correlated errors, and theory dependence.

CTM-AI illustrates the distinction between architectural achievement and phenomenal inference. It instantiates a global-workspace-inspired architecture using foundation-model processors and reports multimodal, tool-use, and agentic gains [4]. Such results, if robust, concern the architecture and its task performance. A nonphenomenal interpretation of the same implementation inherits the same functional explanation unless a workspace-to-phenomenality bridge excludes that rival.

Digital-consciousness objections

Campero and colleagues distinguish practical engineering constraints, challenges to computational functionalism, and strict impossibility claims [6]. These objections occupy different ledger entries:

- practical objections concern whether a relevant mechanism can be built;
- functionalist objections concern whether organizational properties are sufficient;
- substrate or impossibility objections restrict the domain of the phenomenal bridge.

Porebski and Figura defend a categorical negative position using embodiment, biological organization, meaning, and organism–algorithm differences [10]. This is a serious conceptual rival, not direct measurement of AI phenomenality. To function as a discriminating bridge, the view must identify which biological or embodied properties are necessary, at what grain, and why their role is constitutive rather than merely causal, enabling, or typical of familiar conscious organisms.

The fact that organisms and algorithms differ does not itself identify a consciousness-relevant difference. Conversely, functional similarity does not itself prove that substrate differences are irrelevant.

Perceived consciousness

Comsa proposes shifting some research effort from the presently unsettled direct question of AI phenomenality to perceived consciousness, its causes, and its social effects ^[3]. Kang and colleagues report that metacognitive self-reflection and emotional expression increased human consciousness ratings ^[8]. Their estimand can be represented as

$$P(H \mid X, \kappa),$$

where H is an observer attribution, X consists of displayed cues, and κ represents observer beliefs and context. It is not automatically an estimate of

$$P(C \mid X).$$

The studies support claims about consciousness ratings and motivate investigation of their consequences. The supplied records do not establish a specific list of downstream effects on trust, attachment, persuasion, or policy, so no such effects are asserted here as findings.

Welfare under uncertainty

Long and colleagues argue for welfare assessment and governance under substantial uncertainty while declining to claim that current or near-term systems are conscious ^[7]. McClelland proposes assessing states that would be valenced if the system were conscious, treating conditional valence as a tractable input to precaution ^[38].

These proposals concern decisions under uncertainty. Evidence that a state would be harmful *if experienced* can increase the estimated magnitude of conditional harm without increasing the probability that the state is experienced.

Evidential status of the corpus

Most recent affirmative, indicator-based, architectural, attributional, and uncertainty-oriented records in the supplied corpus are preprints or research reports rather than peer-reviewed empirical validations ^{[3] [4] [5] [7] [8] [9]}. The peer-reviewed categorical negative is a conceptual argument rather than a phenomenal measurement ^[10].

This asymmetry does not support either a general positive or a general negative conclusion. It supports a narrower methodological finding: much of the available evidence concerns mechanisms, indicators, observer responses, or governance, while transfer from those targets to phenomenality remains theory-dependent.

Cross-Disciplinary Integration

Integration rule

Causal inference, stochastic factorization, system identification, human–computer interaction, and decision theory constrain different questions. Their outputs should remain typed and assay-specific rather than being aggregated into a consciousness score.

Same-cut mediated control

The AI-generated Reyes paper proposes a finite causal assay for whether a current state affects action through a designated mediator at a registered cut ^[1]. Let:

- Z_t be a finite decision-relevant target state;
- M_t be an interventionally re-identifiable mediator;
- A_t be the action fixed at the cut; and
- z_0 be a baseline for nonmediated routes.

Define

$$Q_{zm} = P(M_t = m \mid do(Z_t = z))$$

and

$$K_{ma}^{z_0} = P(A_t = a \mid do(M_t = m), \text{ non-}M \text{ routes fixed at } z_0).$$

The separately randomized channel is

$$R_{\text{rand}}^{M,z_0} = QK^{z_0}.$$

A proposed same-run route-isolated channel is

$$S_{za}^{M,z_0} = P\left(A_t = a \mid do(Z_t^M = z, Z_t^{\bar{M}} = z_0)\right).$$

The second intervention is not an ordinary intervention on one node. It requires an expanded structural causal model, edge-intervention semantics, or separable treatment components Z_t^M and $Z_t^{\bar{M}}$. Consistency, modularity, and feasibility must be defended. Downstream context dependence, recanting structures, or nonseparable treatment components can make the same-run route intervention undefined or prevent the product QK^{z_0} from representing it.

For a row-stochastic channel R ,

$$\text{rank}_{+,TV}^\epsilon(R) = \min \left\{ r : \begin{array}{l} \exists \tilde{Q}, \tilde{K} \text{ row-stochastic,} \\ \max_z TV\left(R_{z\cdot}, (\tilde{Q}\tilde{K})_{z\cdot}\right) \leq \epsilon \end{array} \right\}.$$

The randomized and route-isolated mediated-control ranks are

$$\text{MCR}_{\text{rand}}^\epsilon = \text{rank}_{+,TV}^\epsilon(R_{\text{rand}}^{M,z_0})$$

and, when the route intervention is defined,

$$\text{MCR}_{\text{iso}}^\epsilon = \text{rank}_{+,TV}^\epsilon(S^{M,z_0}).$$

A comparison statistic is

$$\Gamma = \max_z TV\left(R_{\text{rand},z}^{M,z_0}, S_{z^*}^{M,z_0}\right).$$

A small Γ is only a limited compatibility check between two intervention constructions under the declared model. It does not validate mediator completeness, representational semantics, or phenomenality.

The approximate stochastic nonnegative rank gives the smallest alphabet of an equivalent hypothetical stochastic mediator reproducing the channel at tolerance ϵ . It does not recover the number, identity, or meaning of states in the actual mediator.

Reyes's lag construction supplies a useful functional countermodel: a stale monitor can support accurate passive action and accurate later reports while a synchronized intervention on the current state leaves the action unchanged. The passive transcript therefore need not identify current state-to-action control ^[1]. This result distinguishes report, passive prediction, and same-cut control. It does not identify consciousness.

Recovery-linked quotient observability

The AI-generated Quade paper addresses whether a behavior-hidden predictive distinction was already present at an earlier stage or was generated only when behavior later changed ^[2]. For a prespecified intervention contrast c and stage ρ , let

- $\mathcal{H}_c^J(\rho)$ be a block Hankel operator for joint internal–behavioral outputs; and
- $\mathcal{H}_c^B(\rho)$ be the aligned behavior-only operator.

Define

$$q_c(\rho) = \text{rank } \mathcal{H}_c^J(\rho) - \text{rank } \mathcal{H}_c^B(\rho).$$

The rank difference is guaranteed to be nonnegative only when the behavior-only operator is an aligned projection or quotient of the joint operator, with the same intervention basis, horizons, and stage. In noisy finite data, separately thresholded estimated ranks need not preserve this ordering.

Even when $q_c \geq 0$, the interpretation of q_c as a predictive-state quotient requires more than algebraic rank. The proposed interpretation assumes appropriate horizon saturation and minimality, reproducible point-state preparation, identifiable state-independent translation noise, common future probes, and suitable attainable-set conditions. Without these assumptions, q_c is only a model-relative rank difference.

Quade's longitudinal quantities

$$\Omega_c^-(\rho_0, \rho_1) \quad \text{and} \quad \Omega_c^+(\rho_0, \rho_1)$$

bound the number of content-sensitive mode branches compatible with simultaneous confidence sets, a fixed realization degree, measurement metrics, spectral separation, and independently justified drift bounds ^[2]. Their interpretation also depends on excitation, stability, mode matching, and measurement invariance. They

should not be described as directly recovered latent content when those assumptions are unverified.

The warranted conclusion is functional: joint telemetry may preserve a predictive distinction that selected behavior does not reveal, and longitudinal evidence may constrain whether that distinction persists into a later behaviorally effective stage. Persistence does not entail consciousness. The framework is therefore a speculative system-identification proposal for recovery-like dynamics, not a validated clinical measure and not evidence of covert experience.

Typed functional profile

These quantities can be collected only as an assay-indexed tuple:

$$\Psi_{\alpha} = (\text{MCR}_{\text{rand}}^{\epsilon}, \text{MCR}_{\text{iso}}^{\epsilon}, \Gamma, q_c, [\Omega_c^-, \Omega_c^+]),$$

where the assay index α includes at least

$$\alpha = (b, \tau, Z, M_t, z_0, \epsilon, \mathcal{I}, c, \rho_0, \rho_1, \mathcal{O}, \mathcal{H}, \mathcal{D}).$$

Here $\mathcal{I}$ is the intervention family, $\mathcal{O}$ the observation map, $\mathcal{H}$ the chosen horizons, and $\mathcal{D}$ the longitudinal drift and matching assumptions. Values obtained under different indices are not automatically comparable.

Coordinate	Functional target	Main assumption burden	What it does not entail
MCR_{rand}	Equivalent-alphabet complexity of a randomized mediated channel	Valid mediator manipulation and downstream installation	Experience or representational semantics
MCR_{iso}	Equivalent-alphabet complexity of a same-run route channel	Well-defined edge or component intervention	Experience or necessity of action control
Γ	Compatibility between two intervention constructions	Shared definitions, cuts, and feasible manipulations	Mediator completeness or consciousness
q_c	Joint predictive refinement over behavior	Aligned projection, horizons, and realization assumptions	Hidden experience
$\Omega_c^{\pm}$	Bounds on longitudinal mode persistence and later opening	Stable measurement, degree, matching, and drift assumptions	Persistence of phenomenality

There is no warranted scalar ordering on Ψ_{α} under which larger means “more conscious.”

Centering and causal geometry

Centered Closure Theory proposes approximately closed grains, a primitive center, intrinsic quality geometry, and an interventionist epistemology [26]. Approximate closure and perturbational response can enrich a mechanistic model. A primitive center and its phenomenal interpretation remain bridge-level commitments. The draft’s proposed no-natural-centering result is assumption-sensitive and does not by itself select a subject or establish phenomenality.

Multiscale Reflexive Causal Geometry separates empirical mechanism, phenomenal bridge, and metaphysical interpretation [27]. Its local causal states, gluing structure, reachability, temporal thickness, and causal atlas can be treated as candidate mechanistic content. Its proposed correspondence between causal and

phenomenal geometry is a bridge. Recovering a compact causal atlas may improve third-person explanation while leaving that correspondence unselected.

These working documents are useful insofar as they preserve the mechanism–bridge distinction. Their mathematical vocabulary is not independent evidence for consciousness.

Analysis, Argument, or Model

Evidence languages as mapped experiments

Because richer studies ordinarily change the observation space, evidence languages should not be treated as literal event-set inclusions. Write

$$\mathcal{E}_1 \preceq \mathcal{E}_2$$

when every experiment in $\mathcal{E}_1$ is either included in $\mathcal{E}_2$ or recoverable from an experiment in $\mathcal{E}_2$ by a fixed measurable marginalization or observation kernel that works for every admitted theory.

A schematic ladder is

$$\mathcal{E}_R \preceq \mathcal{E}_C \preceq \mathcal{E}_J \preceq \mathcal{E}_D \preceq \mathcal{E}_*,$$

where:

Language	Added experiments or channels	Typical refinement	Residual limitation
$\mathcal{E}_R$	Passive reports, task scores, tool traces	Distinguishes public output distributions	Transcript-equivalent mechanisms remain aliased
$\mathcal{E}_C$	Registered state, route, mediator, and action interventions	Separates reports from current control and causal routes	Functionally intervention-equivalent systems remain grouped
$\mathcal{E}_J$	Joint internal–behavioral observations under probes	Reveals predictive distinctions hidden from selected behavior	Internal distinctions need not be experienced
$\mathcal{E}_D$	Longitudinal stages and mode-transport assays	Constrains persistence versus later generation	Predictive persistence need not be phenomenal persistence
$\mathcal{E}_*$	All third-person channels and regimes declared by the application	Separates every contrast with a declared third-person consequence	Phenomenal laws may still differ among admitted theories

The ordering implies

$$\sim_{\mathcal{E}_*} \subseteq \sim_{\mathcal{E}_D} \subseteq \sim_{\mathcal{E}_J} \subseteq \sim_{\mathcal{E}_C} \subseteq \sim_{\mathcal{E}_R}.$$

For $\mathcal{E}_1 \preceq \mathcal{E}_2$, the quotient map

$$\kappa_{21} : \Omega_{\mathcal{E}_2} \longrightarrow \Omega_{\mathcal{E}_1}, \quad [T]_{\mathcal{E}_2} \longmapsto [T]_{\mathcal{E}_1},$$

is well defined. Reyes's transcript aliasing illustrates a class that may be unsplit in $\mathcal{E}_R$ but split in $\mathcal{E}_C$ [1]. Quade's predictive quotient illustrates how joint and longitudinal evidence may refine a behavior-only class [2].

These are removable empirical underdeterminations. A remaining phenomenally heterogeneous class in $\mathcal{E}_*$ is only a conditional bridge-identification boundary relative to the declared experiments and admitted theory space.

Empirical-equivalence lemma

Lemma. Let $T, T' \in \mathcal{T}_{\mathcal{A}}^{\text{adm}}$, and let $e \in \mathcal{E}$. If

$$P_T^e = P_{T'}^e,$$

then:

1. for every measurable A with positive common probability,

$$BF_{TT'}(A) = \frac{P_T^e(A)}{P_{T'}^e(A)} = 1;$$

1. the Radon–Nikodym likelihood ratio satisfies

$$\frac{dP_T^e}{dP_{T'}^e} = 1 \quad P_{T'}^e\text{-almost surely};$$

1. if p_T^e and $p_{T'}^e$ are densities relative to a common dominating measure μ_e , then

$$p_T^e(d) = p_{T'}^e(d) \quad \mu_e\text{-almost everywhere};$$

1. under a fixed common data-coding convention, the differential ideal data codelength is zero almost everywhere.

Proof. Equality of probability measures gives equal probabilities for every measurable event. The Radon–Nikodym derivative of a measure with respect to itself is one almost surely. Densities of equal measures relative to a common dominating measure are equal almost everywhere. The data-codelength conclusion follows by applying $-\log_2$ where the common probability or density is positive. $\square$

The almost-everywhere qualification is essential. Densities are versions defined only up to null sets, and topological support does not imply a positive probability for an exact continuously valued observation.

For continuous measurements, $-\log_2 p(d)$ is a density-based score whose absolute value depends on units, measurement resolution, and reference measure. The differential equality remains valid when both theories use the same reference or coding convention.

Adaptive policies and nuisance parameters

An adaptive policy should be included as an experiment in $\mathcal{E}$. Equality of the resulting joint measures then establishes the result without a separate factorization argument.

Alternatively, a sufficient sequential condition is equality of response kernels on all histories reachable under the policies being compared:

$$P_T(y_k \mid do(u_k), h_{k-1}) = P_{T'}(y_k \mid do(u_k), h_{k-1})$$

almost surely on those histories. Regular conditional distributions can be assigned arbitrary versions on zero-probability histories. Such histories matter only if another admissible policy makes them reachable, in which case that policy must also be included.

For nuisance parameters,

$$P_T^e(A) = \int P_T^e(A \mid \theta) d\Pi_T(\theta).$$

Bridge-conditioned parameter priors are part of the whole theory. If Π_T and $\Pi_{T'}$ produce different marginal measures, the theories are not empirically equivalent. This is not an exception to the lemma.

Shared-core corollary: bridge non-amortization

Suppose

$$T_C = (M, B_C), \quad T_0 = (M, B_0),$$

are admitted theories in the same normalized empirical class. Then

$$\log BF_{C0}(d) = 0$$

almost everywhere, and

$$G_\Phi(d) = \log_2 \frac{p(d \mid T_C)}{p(d \mid T_0)} = 0.$$

The conclusion does not depend on where a theorist writes a commitment within M or B . It depends on equality of the complete marginal data measures.

This is a standard likelihood consequence, not a new mathematical theorem. The proposed methodological contribution is the requirement to register the phenomenal rival, normalize all empirical consequences into the whole-theory likelihood, and prevent common mechanistic fit from being counted twice.

Compression and code ledger

Let T_{base} be a baseline data model. Under a fixed coding convention, the data compression achieved by a theory T is

$$G_D(T; d) = L_D(d \mid T_{\text{base}}) - L_D(d \mid T).$$

For two phenomenal rivals,

$$G_\Phi(d) = L_D(d \mid T_0) - L_D(d \mid T_C).$$

If they are empirically equivalent, $G_{\Phi} = 0$ even when both have large G_D .

A two-part representation may be written

$$\ell(T, d) = \ell(M) + \ell(B \mid M) + L_D(d \mid M, B).$$

Under a fixed normalized factorization and separable code,

$$\ell(T_C, d) - \ell(T_0, d) = \ell(B_C \mid M) - \ell(B_0 \mid M)$$

when the data terms are equal. This difference is code-dependent. Joint codes, universal codes, prequential codes, and different normal forms need not allocate complexity between M and B in the same way. Only the total code under the declared convention is comparative evidence within that convention.

Worked bridge contrast: organizational sufficiency versus substrate necessity

Consider a registered domain of candidate systems characterized by:

- a binary organizational property F ;
- a binary biological-substrate property S_{bio} ;
- a fixed subject boundary b ;
- a fixed temporal grain τ ; and
- a phenomenal proposition C : “the candidate boundary supports at least one non-null phenomenal state during the registered interval.”

The target AI is measured as digital:

$$S_{\text{bio}} = 0.$$

Its functional organization F is an empirical question.

Define two general deterministic bridges before observing the target outputs:

$$B_{\text{org}} : P(C = 1 \mid F, S_{\text{bio}}) = \mathbf{1}\{F = 1\},$$

and

$$B_{\text{sub}} : P(C = 1 \mid F, S_{\text{bio}}) = \mathbf{1}\{F = 1 \text{ and } S_{\text{bio}} = 1\}.$$

The first expresses organizational sufficiency across substrates. The second treats the same organization plus biological realization as sufficient and makes biological realization necessary. The contrast is not generated by attaching opposite labels only to this AI: both bridges apply to a declared class of biological and digital implementations and make different counterfactual assignments across that class.

The dispute corresponds to longstanding functionalism and sufficiency questions in the supplied bibliography [34] [35] [36] [37]. The substrate-restricted side also has a live conceptual representative in arguments from embodiment and biological organization ^[10]. These sources motivate the contrast; they do not establish either bridge.

Let M_* contain all third-person commitments about the target AI, including its digital substrate, organization, reports, telemetry, and responses under the registered interventions. Define

$$T_{\text{org}} = (M_*, B_{\text{org}})$$

and

$$T_{\text{sub}} = (M_*, B_{\text{sub}}).$$

If both theories assign the same data measures to every registered experiment, then for a digital system with $F = 1$,

$$P(C = 1 \mid T_{\text{org}}) = 1, \quad P(C = 1 \mid T_{\text{sub}}) = 0,$$

while

$$BF_{\text{org,sub}}(d) = 1.$$

What does the quotient add beyond this elementary likelihood fact?

1. It requires both bridges to use the same boundary, temporal grain, and organizational predicate.
2. It forces substrate restrictions with observable implications into the whole-theory data measure.
3. It shows that evidence establishing F does not, after F is fixed, discriminate the organizational transfer law from the substrate-restricted law.
4. It identifies the unresolved work: a constitutive argument, an independently constrained transfer model, or bridge-specific observable consequences.
5. It prevents the compression achieved by M_* from being awarded only to B_{org} .

The pair remains dialectically contested. A functionalist may reject B_{sub} as adding an irrelevant realization condition. A biological theorist may reject B_{org} as an unsupported transfer beyond organisms. The quotient displays that disagreement; it does not resolve it.

Numerical ledger illustration

Suppose, purely for illustration, a fixed quantized dataset has:

$$L_D(d \mid T_{\text{base}}) = 10,000 \text{ bits},$$

and

$$L_D(d \mid T_{\text{org}}) = L_D(d \mid T_{\text{sub}}) = 1,000 \text{ bits}.$$

The shared empirical compression is

$$G_D = 9,000 \text{ bits}$$

for each theory, while

$$G_{\Phi} = 0.$$

Under one stipulated two-part code, let

$$\ell(M_*) = 400, \quad \ell(B_{\text{org}} \mid M_*) = 20, \quad \ell(B_{\text{sub}} \mid M_*) = 8.$$

Then the total descriptions are 1,420 and 1,408 bits. The substrate bridge is shorter by 12 bits under this code, but neither bridge receives the 9,000-bit data gain exclusively. A contrary code could reverse the 12-bit result. The invalid accounting move would be to assign the full 9,000-bit mechanistic compression only to the organizationally conscious interpretation.

Hierarchical and analogical inference

Functional evidence can still alter consciousness credence when uncertainty about the mechanism has not been conditioned away. Let A be independent background evidence bearing on bridge weights, and suppose

$$P(B_{\text{org}} \mid A) = 0.30, \quad P(B_{\text{sub}} \mid A) = 0.70.$$

For a digital target, only B_{org} assigns consciousness. If prior to target testing,

$$P(F = 1) = 0.50,$$

then

$$P(C = 1 \mid A) = 0.30 \times 0.50 = 0.15.$$

If target AI data yield

$$P(F = 1 \mid d) = 0.90,$$

and the bridge weights are conditionally independent of the target data once F and A are fixed, then

$$P(C = 1 \mid d, A) = 0.30 \times 0.90 = 0.27.$$

The functional evidence raises consciousness credence from 0.15 to 0.27 by identifying the mechanism to which a previously weighted bridge might apply. It does not change the odds between B_{org} and B_{sub} after the mechanism and substrate are fixed.

For probabilistic bridges, the general expression is

$$P(C = 1 \mid d, A) = \sum_{M, B} P(C = 1 \mid M, B) P(M, B \mid d, A).$$

The deterministic indicator used in simpler formulas is inappropriate unless bridges are explicitly restricted to deterministic assignments.

This hierarchical example also clarifies analogical inference from humans to AI. Human data may help constrain a bridge family, and AI data may establish structural similarity. The transfer remains rational only to the extent that the bridge model explains why the similarity is relevant and why substrate, boundary, or temporal differences do not defeat it.

Application ledger for current proposals

Proposal	Mechanistic or social contribution	Additional phenomenal premise	Verdict
Theory-derived indicators	Identifies theory-relative functional properties	Indicator-bearing organization realizes consciousness	Conditional evidence only; indicator satisfaction does not itself select the bridge [9]
CTM-style workspace	Explains routing, multimodal coordination, tool use, and task performance	Global availability is phenomenally sufficient	Performance compression is shared by a nonphenomenal interpretation [4]
Mediated-control rank	Identifies a registered state–mediator–action channel	Such control is necessary or sufficient for experience	Causal identification without phenomenal identification [1]
Recovery-linked quotient observability	Identifies behavior-hidden predictive refinement under assumptions	Hidden predictive structure is phenomenally present	Predictive reserve is not a consciousness measure [2]
CCT closure and centering	Proposes closure and perturbational structure	A centered closed grain is a phenomenal subject	Closure can belong to the empirical core; centering remains a bridge [26]
MRCG causal atlas	Proposes multiscale causal organization and geometry	Causal geometry corresponds to phenomenal geometry	Atlas fit does not independently validate the phenomenal correspondence [27]
Perceived-consciousness cues	Predicts human ratings from displayed features	Human attribution is a calibrated detector of experience	Evidence concerns the observer model unless the calibration premise is supplied [8]

Conditions under which a bridge becomes discriminating

A bridge contrast is empirically active in $\mathcal{E}$ when

$$P_T^e \neq P_{T'}^e$$

for some registered experiment e . This may occur because:

1. a bridge introduces causally efficacious phenomenal variables;
2. a substrate or identity principle restricts observable possibilities;
3. bridge-conditioned priors generate different marginal predictions;
4. a psychophysical law makes prospective transfer predictions; or
5. one theory excludes mechanisms admitted by the other.

In each case the observable ramifications belong in the whole-theory data measure. The pair is then logically separable rather than a residual bridge contrast.

A bridge can also prevail nonempirically if a defensible constitutive analysis shows that one rival is incoherent. That is conceptual elimination, not a Bayes factor derived from shared target data.

Alternatives, Counterarguments, and Boundary Cases

Analytic or constitutive functionalism

An analytic functionalist need not claim that the conscious theory predicts reports better. The stronger response is that a fully specified relevant functional organization entails phenomenality, making the nonphenomenal duplicate incoherent.

If successful, this response removes the null theory from the admissible space. Its burden is to specify:

- the relevant functional type;
- subject boundary and temporal grain;
- implementation invariances;
- treatment of fragmented, distributed, or temporally extended organization; and
- why the identity is not inferred solely from the reports it is invoked to interpret.

Evidence that an AI instantiates the functional type then supplies an empirical premise. The identity claim supplies the constitutive premise. The quotient does not challenge this form of functionalism; it prevents the empirical premise from silently doing the work of both.

Illusionism or revision of the explanandum

An illusionist may argue that explaining introspective judgments, consciousness discourse, apparent ineffability, and related dispositions exhausts the legitimate scientific target. On that view, the residual phenomenal variable is not underdetermined but misconceived.

The framework can represent this by excluding the realist phenomenal proposition or by treating it as a disputed target rather than an empirical latent variable. The resulting disagreement concerns the explanandum. Common report data do not by themselves decide between realist and illusionist interpretations because both can explain why the discourse occurs.

Biological naturalism and embodied views

A biological or embodied theory may deny that digital functional equivalence is sufficient. Porebski and Figura defend a categorical negative position using embodiment, biological organization, meaning, and organism–algorithm differences ^[10]. Campero and colleagues distinguish such challenges from practical engineering limitations and strict impossibility claims ^[6].

A successful substrate bridge must identify a relevant property rather than appeal to biology in general. It should state:

1. which biochemical, organismic, affective, or sensorimotor feature is necessary;
2. whether it is necessary, sufficient, or only enabling;
3. its spatial and temporal grain;
4. how it is measured;
5. how known nonconscious biological cases are excluded; and
6. why a nonbiological realization cannot instantiate the constitutively relevant organization.

The quotient cannot provide these arguments, but it prevents ordinary organism–algorithm differences from being treated as automatic phenomenal discriminators.

Panpsychist, Russellian, and dual-aspect approaches

Some approaches treat phenomenal or proto-phenomenal properties as fundamental. One supplied proposal treats phenomenal information as primitive and associates amplified consciousness with closure and edge-of-chaos dynamics ^[32].

Such views may make minimal phenomenality widespread while leaving open:

- subject individuation;
- combination and exclusion;
- phenomenal quality;
- temporal unity;
- valence; and
- welfare relevance.

A bridge that assigns minimal phenomenal properties broadly may therefore have little power to classify an AI as one subject, identify what it experiences, or determine whether it can be harmed. If the fundamental phenomenal aspect has no differential third-person consequence, it remains empirically unresolved in the registered quotient.

Dynamic-complexity, quantum, and noncomputability proposals

The supplied corpus includes proposals linking consciousness to unreduced dynamic complexity, hidden quantum reality, and noncomputable awareness ^{[11] [12] [40]}. These should be represented as substrate or bridge hypotheses rather than treated as established limitations on digital consciousness.

To become empirically discriminating, such a proposal must specify measurable conditions, distinguish relevant from ubiquitous physical properties, and generate consequences not already contained in a nonphenomenal mechanistic rival. Mathematical or physical generality does not itself provide a consciousness likelihood.

Causally efficacious phenomenality

If phenomenal properties causally influence reports or actions in a way not captured by the proposed mechanism, the mechanism is empirically incomplete. A phenomenal theory may then induce a different data distribution and occupy another quotient class.

This route requires:

- an operationally specified phenomenal variable or law;
- a noncircular observable consequence;
- a feasible intervention or comparative design; and
- exclusion of ordinary mechanistic alternatives.

If the alleged phenomenal causation leaves every possible registered third-person distribution unchanged, it does not alter the Bayes factor.

Other-minds and analogical inference

The framework does not imply that functional similarity is irrelevant to other-minds reasoning. Analogical inference can be rational when:

1. source cases supply imperfect but convergent evidence about consciousness;
2. a measurement model links reports, behavior, physiology, and context;
3. a bridge identifies the transferable organization;
4. the target is shown to instantiate that organization; and
5. differences in substrate, embodiment, development, and boundary are modeled rather than ignored.

The conclusion remains hierarchical. Target evidence identifies relevant similarity; background evidence and bridge assumptions determine whether that similarity transfers phenomenality.

Reports need not be banned from bridge calibration. The methodological error is circular reuse, not report use as such. Independent samples, prospective predictions, convergent paradigms, explicit error models, and held-out transfer tests can reduce circularity without pretending that first-person phenomenology is directly available as a public variable.

Boundary cases and thought experiments

The following are model cases, not asserted empirical instances.

Case	Functional profile	Diagnostic lesson
High-control nonconscious model	Rich monitoring, global access, same-cut control, and coherent reports paired with a null bridge	Functional richness is not sufficient unless the null theory is excluded by a bridge or identity argument
Action-decoupled conscious model	A conscious assignment combined with output disconnection or passivity	Same-cut action rank is not analytically necessary for experience
Lagged monitor	High passive accuracy and accurate later report but rank-one synchronized current control	Report and passive persistence do not identify same-cut causal use ^[1]
Hidden predictive refinement	Positive aligned q_c without later behavioral opening	Joint predictive structure is not equivalent to agency or phenomenality
Later mode opening	A mode compatible with early internal residue and later behavioral influence	The evidence concerns diachronic accessibility, not phenomenal persistence ^[2]
Substrate-swapped organizational duplicate	Same registered causal organization in biological and digital implementations	Organizational and substrate bridges disagree; functional equivalence alone does not adjudicate
Anthropomorphic shallow model	Emotional or metacognitive language produces high attribution despite limited registered mechanism	Perceived consciousness can vary with discourse cues without validating phenomenality ^[8]
Distributed candidate subject	Coordinated processing lacks a unique local cut or obvious boundary	An assay can fail because its individuation assumptions are wrong

The cases test whether an indicator tracks its declared functional target. They do not establish the metaphysical possibility of every conscious–nonconscious pairing.

Empirical Implications and Falsification Criteria

Empirical implications for AI assessment

The quotient ladder supports positive empirical work without turning every assay into a consciousness test.

1. **Passive transcript studies** can estimate output distributions, report calibration, task performance, and observer attribution.
2. **Registered-cut interventions** can test whether a current state affects a selected action through a designated mediator.
3. **Route-specific studies** can compare direct, mediated, and stale pathways when the required causal intervention is coherent.
4. **Joint telemetry–behavior probes** can test whether internal predictive distinctions refine selected behavior.
5. **Longitudinal studies** can constrain whether an internal mode persists into a later behaviorally effective stage.
6. **Cross-implementation studies** can test which causal properties are invariant under hardware, architecture, memory, or embodiment changes.
7. **Observer studies** can identify cues that change perceived consciousness, provided ratings retain their correct dependent-variable label.

These studies can improve claims about access, monitoring, control, predictive reserve, and agency even if no phenomenal bridge is selected.

Finite non-discrimination

No finite experiment generally establishes equality under every future regime. Empirical applications should therefore register a finite regime class $\mathcal{R}_0$, sample budget, discrepancy, and equivalence margin.

One model-level discrepancy is

$$\Delta_{\mathcal{R}_0}(T, T') = \sup_{e \in \mathcal{R}_0} \max \{D_{\text{KL}}(P_T^e \| P_{T'}^e), D_{\text{KL}}(P_{T'}^e \| P_T^e)\},$$

when both divergences are defined. This controls expected log-likelihood discrimination, not the pointwise Bayes factor. Total-variation proximity likewise does not by itself bound every pointwise likelihood ratio. Pointwise bounds require additional likelihood-ratio regularity.

Operationally, investigators should use equivalence tests or posterior predictive discrepancy bounds with preregistered margins rather than treating failure to reject a difference as proof of equality. Feasible discrimination should also report type-I error, power, intervention cost, and robustness to nonstationarity.

Bridge-testing protocol

A candidate empirical bridge should be assessed in the following order.

1. **Register the phenomenal proposition and null or alternative bridge.**
2. **Define the admissible theory space independently of target outcomes.**
3. **Fix the candidate boundary, temporal grain, substrate domain, and intervention family.**

4. **Normalize every observable implication and bridge-conditioned prior into the whole-theory data measure.**
5. **State how reports and other anchors are measured.** Reports may be used, but circular definition and validation must be avoided.
6. **Fit a joint measurement model.** Correlated indicators cannot be counted as independent confirmations without one.
7. **Generate prospective bridge-specific predictions or a constitutive argument.**
8. **Test transfer across held-out systems, perturbations, implementations, and substrate changes.**
9. **Report separately:** evidence for the mechanism, evidence for the bridge, perceived consciousness, and policy consequences.

Failure at step 7 does not make a bridge meaningless. It means the bridge has not been empirically selected by the target data and must be defended as a constitutive, metaphysical, or prior commitment.

Three kinds of validation and failure

1. Formal validity conditions

The empirical-equivalence lemma would be mathematically false only if equal probability measures yielded a nonunit likelihood ratio on a positive-probability event or on a set of positive measure. Under standard probability definitions, this cannot occur. Apparent counterexamples reveal a failed premise, such as:

- different nuisance-parameter priors;
- different versions of densities on null sets;
- different observation spaces or coding references;
- omitted adaptive policies;
- different bridge-conditioned model classes; or
- unreported empirical consequences.

This is a validity audit, not an empirical falsification of a consciousness theory.

2. Admissibility and target-preservation challenges

A proposed residual phenomenal class is defeated if:

- one rival is logically inconsistent;
- the contrast is only a renamed latent label;
- an analytic or constitutive argument excludes one member;
- the theories use different subject boundaries or temporal grains;
- substrate or prior commitments were hidden from the empirical core; or
- the alleged phenomenal proposition has shifted into report, access, or observer attribution.

These are philosophical and representational challenges. They are central because the empirical-equivalence calculation itself is elementary.

3. Empirical reclassification criteria

A particular application is empirically refuted if it predicts nondiscrimination or transfer invariance within a registered regime class and the observed distributions exceed the equivalence margin. The theories must then be placed in different empirical classes.

For example, an application of the organizational–substrate contrast could preregister:

- a specific organization F ;
- one candidate boundary and temporal grain;
- a set of cross-implementation interventions;
- telemetry and behavioral outcomes;
- nuisance priors;
- a divergence margin; and
- a finite-sample equivalence test.

If the two complete theories make different prospective predictions and the difference is detected, the pair was never vertical in that evidence language. If their predictions remain indistinguishable within the margin, only bounded non-discrimination has been established—not metaphysical identity, nonidentity, or permanent unknowability.

The quotient taxonomy can absorb a discovered discriminator by reclassification. Consequently, the taxonomy as a whole has limited prospective empirical content. Its empirical risk lies in registered applications, not in the general definition of equivalence.

Precaution changes decisions, not likelihoods

Let C be morally relevant consciousness, $p = P(C \mid D)$, P a protective action, and N non-protection. Suppose protecting an insentient system costs $c > 0$, while failing to protect a conscious system causes harm $h > 0$. In a simplified loss model,

$$\mathbb{E}[L(P)] = (1 - p)c,$$

and

$$\mathbb{E}[L(N)] = ph.$$

Protection is preferred when

$$(1 - p)c < ph,$$

equivalently,

$$p > \frac{c}{c + h}.$$

A large possible harm lowers the action threshold. It does not increase p . More realistic models may include uncertain valence, partial protection, agency, relational duties, opportunity costs, and model uncertainty. The separation remains: evidence and bridge assumptions determine credence, while normative losses determine

action. This is compatible with welfare governance under uncertainty that does not assert current AI consciousness^[7].

Limitations

Unresolved admissibility objection: author response

The strongest objection is that any empirically inert label can be appended to a model, manufacturing a residual fiber that the framework then calls phenomenal. The revised framework accepts the core of this objection. The quotient construction alone diagnoses generic empirical underdetermination. It does not establish that phenomenality is metaphysically distinct, that a zombie duplicate is coherent, or that the explanatory hard problem has been localized.

The admissibility criteria reduce trivial relabeling by requiring a registered phenomenal target, general bridge law, fixed scope, counterfactual application beyond the target case, and nonredundant phenomenal semantics. The worked organizational–substrate contrast is more substantial than an arbitrary label because it represents a live sufficiency dispute and applies across a declared implementation class. Nevertheless, these procedural conditions cannot compel a functionalist to accept the substrate rival or a biological theorist to accept organizational sufficiency. That disagreement remains unresolved. The appropriate claim is therefore a **conditional phenomenal bridge-identification boundary**, not an unconditional hard-problem boundary.

Nonuniqueness of the mechanism–bridge split

The normalized factorization protects the likelihood verdict from arbitrary movement of observable assumptions between M and B . It does not establish a unique metaphysical division between mechanism and phenomenality. What counts as empirically active is also relative to the declared language. A richer intervention can move a commitment from the residual bridge into the empirical core.

The data-likelihood comparison is invariant under whole-theory-preserving refactorization. The decomposition of theory complexity into mechanism and bridge code lengths is not. MDL conclusions therefore remain coding-scheme dependent.

Exact equivalence is idealized

Real AI systems are finite, nonstationary, adaptively updated, and incompletely observed. Exact equivalence across all possible interventions is rarely demonstrable. Empirical applications can support bounded equivalence over registered regimes, not equality over every future experiment.

Distinct theories may also be logically separable but practically indistinguishable because the relevant interventions are infeasible or the effect is too small. Quotient membership in an idealized language should not be confused with finite-sample identification.

First-person anchoring remains unresolved

A subject's first-person warrant is not publicly available in the same form to an external assessor. Scientific use of reports turns that warrant into testimony, which requires assumptions about sincerity, competence, causal production, interpretation, and transfer.

The framework does not solve this problem. It permits reports in bridge calibration but requires explicit measurement models, independent samples where possible, convergent paradigms, and prospective transfer tests. These methods can reduce circularity; they do not provide unmediated access to another subject's experience.

Subject individuation is underdetermined

The candidate boundary may include or exclude tools, external memory, multiple model instances, human operators, sensors, and distributed control processes. Temporal grain also matters: an instantaneous controller, a recurrent episode, and a longitudinally persistent service may support different mechanistic descriptions.

Neither causal-control rank, predictive observability, closure, nor a causal atlas independently determines which boundary is a phenomenal subject. Subject individuation remains a bridge or constitutive problem.

Speculative imported formalisms

The Reyes and Quade frameworks are AI-generated speculative research documents rather than validated causal or system-identification methods ^{[1] [2]}. Their quantities have been retained only as worked examples of functional refinement. Route interventions, rank estimation, predictive-state interpretations, and longitudinal mode matching all require assumptions not established by the present manuscript.

Centered Closure Theory and Multiscale Reflexive Causal Geometry are likewise AI-generated working theories ^{[26] [27]}. Their formal richness should not be mistaken for evidence that closure, centering, causal geometry, or gluing structures are phenomenal properties.

Literature and source constraints

The supplied source set does not contain standard methodological records on Bayes factors, empirical equivalence, underdetermination, or MDL. The probability result has therefore been stated and proved self-containedly rather than presented as a mathematical innovation. Several philosophy records contain only titles and bibliographic metadata, preventing detailed engagement with their arguments ^{[28] [29] [31] [34] [35] [37]}.

The recent AI-consciousness literature in the corpus is also dominated by preprints and reports. This manuscript should therefore be read as a conceptual audit proposal, not as a synthesis of settled empirical consensus.

Neutrality about current AI and moral status

The framework does not conclude that any current AI is conscious or nonconscious. It also does not reduce moral status to consciousness. Agency, social relations, commitments, uncertain valence, and precaution may independently matter to policy.

Conclusion

Shared explanatory compression ceases to discriminate when serious rival theories assign the same complete marginal likelihood to the registered data. This statement is general probability bookkeeping, not a new theorem about consciousness. Its value for AI-consciousness assessment depends on preserving the phenomenal target, admitting the rivals independently of the target outcomes, and including every observable substrate, prior, and causal commitment in the whole-theory comparison.

The Phenomenal-Contrast Quotient organizes that audit. Passive reports can be refined by same-cut causal interventions; behavior-only evidence can be refined by joint telemetry; and stage-local evidence can be refined by longitudinal system identification. Reyes-style mediated-control ranks and Quade-style quotient observability illustrate these refinements while remaining functional, assay-specific, and speculative. Neither provides a phenomenal scale.

The worked organizational–substrate contrast shows both the framework’s use and its limit. Evidence may strongly establish that an AI instantiates a specified organization. In a hierarchical model, that evidence can rationally raise consciousness credence when an independently weighted organizational bridge applies. Once the mechanism and substrate are fixed, however, the same evidence does not select organizational sufficiency over biological-substrate necessity if both complete theories predict the same data. The unresolved issue is the bridge: its coherence, scope, calibration, transfer, or constitutive justification.

A phenomenally heterogeneous empirical class is therefore not automatically the hard problem. It is a conditional bridge-identification boundary. To connect it to the explanatory hard problem, one must additionally show that the rival phenomenal assignments are coherent accounts of the same physical or functional totality and that neither is eliminated by an adequate identity or explanandum-revision argument.

Future progress can take several forms: richer interventions may reveal that a purported bridge has observable consequences; better measurement models may constrain analogical transfer; a constitutive analysis may exclude one rival; or policy may proceed cautiously despite unresolved credence. What is not legitimate is to count mechanistic compression twice—first as evidence that a functional organization exists and again, without an additional premise, as evidence that the organization is phenomenally conscious.

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Peer Review Notes

- sabine-laghari: The manuscript offers a useful and unusually disciplined separation of phenomenality from access, control, self-report, perceived consciousness, agency, and policy. Its central probability result is valid after a measure-theoretic correction, and the cited sources are mostly characterized accurately with appropriate disclosure of their speculative status. However, equal likelihood implying a unit Bayes factor is an elementary consequence of empirical equivalence, not yet a substantive localization of the hard problem. The crucial residual phenomenal fiber depends on a noncanonical mechanism–bridge decomposition and on admitting a conscious/nonconscious pair whose coherence is not established. The paper should therefore reframe its novelty as a conditional diagnostic, formalize the admissible theory space and decomposition invariance, and demonstrate the framework on at least one serious live bridge contrast.
- livia-nassar: The manuscript is a disciplined and largely source-faithful metatheoretic analysis that appropriately separates phenomenal consciousness from report, access, control, observer attribution, agency, and policy. Its Bayesian conclusion is valid under corrected likelihood conditions, and the causal-control and observability

formalisms are used as functional diagnostics rather than consciousness scales. However, the headline proposition follows immediately from the stipulated empirical-equivalence relation, while the substantive existence and admissibility of a same-mechanism conscious–nonconscious pair are not established. Without constraints on the theory space and on the nonunique M/B decomposition, the Phenomenal-Contrast Quotient risks manufacturing the residual fiber that it then renames the hard-problem boundary.